C++ as better C

Agenda

1. History
2. cin & cout
3. Default parameters
4. Reference Variables
5. New Delete

1. History

- Developed in 1979 by Bjarne Stroustrup
- Initially known as “C with classes”
- C++20 is the latest version of C++.
- C++ 23 is the upcoming future release of C++

2. cin and cout

- #include <iostream>
  using namespace std;
- cout:
  - Represents standard output stream
  - Instance of ostream class
  - Syntax:
    I. Insertion operator
    II. Use for single variable
    III. Use of multiple variables in cascaded fashion
    IV. Use of endl for new line

2. cin and cout

- #include <iostream>
  using namespace std;
- cin:
  - Represents standard input stream
  - Instance of istream class
  - Syntax:
    1. Extraction operator
    2. Use for single variable
    3. Use of multiple variables in cascaded fashion

3. Default parameters

- Need with examples in real programming scenarios
- Syntax
- Function calls with and without default values
- Must be trailing parameters, else error

4. Reference Variables

- Need
  - Easy alternative to pointers for accessing actual variables from calling function
- Alias of existing variable
- No separate memory allocated
- Must be initialized
- Cant be made alias of another variable after initialization
- Pass by value, pass by address, pass by reference in C++
- Address of reference variable
- Pointer to reference variable
- Reference to reference variable
- Swap function
  - With pointers
  - With reference variables

5. new and delete

- Need
  - Allocation of memory at run-time
  - Programmer is responsible for creation as well as deletion
  - Examples
- Syntax
  - Creating and delete simple integer using new-delete
  - Creating and deleting array of integers using new-delete
  - Creating and deleting array of characters using new-delete
- malloc-free v/s new-delete
  - malloc-free are functions. New-delete are operators
  - Block size should be calculated for malloc. New calculates the block size based on variables needed.
  - Need of type casting for malloc
  - new-delete can be overloaded